# Response to Reviewers — GSEV-D-26-00126 (v2)

## Overview

Manuscript GSEV-D-26-00126, “Genetic Parameters and Selection Responses for Alternative Methane Trait Definitions in Pasture-Based Sheep”, Genetics Selection Evolution. Revision due 27 Oct 2026.

We thank the Editor and both reviewers for their careful and constructive reading of our manuscript. Below we reproduce each comment in full followed by our response. In brief, we have (i) re-examined the genetic-parameter analysis in detail and added carbon dioxide as a trait in its own right; (ii) re-estimated the parameters used in the selection index analysis from a trivariate model which includes: methane, metabolic body weight and carbon dioxide, and recalculated the selection index analysis based on these results; and (iii) changed the aim of the selection index analysis, which no longer predicts responses under a breeding index but instead examines how absolute, ratio and residual methane traits behave when included in an index. All changes made in response to the reviewers' comments are highlighted in the revised manuscript. Line numbers given in our responses refer to the revised, highlighted manuscript.

## Editor

*Both reviewers found the paper interesting and well written. They raised however a number of points which need to be addressed before the article can be accepted. I ask the authors to account for all these remarks in a revised version and to reply point-by-point in the cover letter.*

**Response:** We thank the Editor for the opportunity to revise the manuscript. Each comment is reproduced below with a point-by-point response, and the line numbers given with each response show where the corresponding changes are in the revised manuscript.

## Reviewer #1

*This is an interesting paper on a topic of considerable international interest. Specifically, it consists of 3 components. The first is estimation of a wide variety of genetic variance and covariance parameters from a significant industry dataset that includes methane measurements. The second is estimation of genetic progress under a variety of indices, both in production traits and in reduction in methane emissions. The final is the consequence of using different definitions of methane emission. There have been previous investigations of each of these over the past 20 years, but not all three at once in an integrated paper.*

*However, I feel that the paper needs some significant revision before resubmission. For three reasons:*

*• In the first case, one of the genetic parameter estimates conflicts badly with previous published work, which is not commented on, but makes one suspect that there has been a mistake in the analysis.*

*• The second situation is that the traits and selection indices used have not been justified and differ rather dramatically from an economically based farm model where the cost of feed is typically the major input cost.*

*• The final issue is that simplistic use of Smith-Hazel index theory was utilised when in industry the situation is significantly more complicated, but there is no reference in the text to the implications or caveats that this entails.*

*Dealing with each in turn. A substantial part of the paper deals with description of the data set and estimation of genetic parameters. My attention was initially drawn to the CH4 ratio trait, which had a heritability estimate of 0.08(0.02) and a repeatability of 0.09(0.01). A prior paper (Jonker et al, 2018) had estimates in respiration chambers with lambs of 0.17(0.03) and 0.43(0.02) and in lambs in PAC chambers of 0.19(0.04) and 0.27(0.02) and ewes in PAC chambers of 0.25(0.05) and 0.38(0.03). This paper is referenced, but the major discrepancy in the estimates between the two studies is not mentioned or acknowledged. The differing results*

*made me suspect that there is some underlying issue with the analysis itself. Looking at the description more closely, I identified.*

**Response:** We thank the reviewer for raising this. On checking Jonker et al. (2018), we found that their ratio trait is defined as CH4/(CH4 + CO2) on a molar basis (mol/mol), whereas our submission calculated it on a mass basis (g/g), which we have now clarified in the text (lines 170–171). We recalculated the trait on a molar basis, which raises the heritability from 0.10 to 0.13 under a homogeneous residual variance. We separately found that allowing residual variance to differ across contemporary-group-mean classes raises the mass-basis heritability further, to 0.19, closely matching the Jonker et al. (2018) lamb estimate. In light of this, and of our reflection on the overall modelling strategy prompted by the other reviewers' comments (in particular on combining growing and mature animals, addressed below), the revised manuscript now uses the heterogeneous-residual model as the primary analysis for all traits investigated in the present study with the exception of CH4/ADG, CH4/MM, CH4/rumen and RMTADG, which were recorded only in growing animals or on the small CT subset. We believe the units difference and the residual-variance structure explain the great majority of the original discrepancy, and we now discuss this explicitly in the manuscript. We also note that Jonker et al. (2018) data come from a different population (New Zealand lambs and ewes) than our own (Irish pasture-based sheep of multiple breeds), which alone would be expected to produce some difference in heritability estimates. The corresponding repeatability under this model is 0.32 (Table 2). The full re-analysis is described in the Methods, with further supporting detail in Supplementary Material S1. [Revised manuscript, lines 170–171, 214–224, 339–341, 482–491.]

*CH4/MBW is defined on page 8 and then referred to as MI throughout the rest of the text*

**Response:** This has been corrected; the trait is now referred to consistently as CH4/MBW throughout the manuscript.

*The units used for the calculation of CH4 ratio are not explicitly defined are they in grams per day or moles? In fact we don't even have a table of means and units for such things as carbon dioxide emissions. The values are small so a rounding error is possible.*

**Response:** As noted above, the units used to calculate CH4 ratio are now stated explicitly in the Methods: both methane and carbon dioxide are expressed in grams per day (g/day), making CH4 ratio a mass-basis proportion. Units for all traits are now also given in Tables 1 and 2. We have added carbon dioxide as a fully reported trait throughout the paper: a means/SD/range entry in the descriptive-statistics table, its own row in the genetic-parameters table, and its genetic and phenotypic correlations with methane and metabolic body weight, addressing the reviewer's point that no units or descriptive table were previously given for CO2. [Revised manuscript, lines 165–166, 170–171, 293–294.]

*In a paper estimating a comprehensive set of genetic parameters I would expect a formal tabulation of the significance of the fixed effects and covariates in the model and even where justified the proportion of variation explained by the factors. For instance what proportion of the variation do breed estimates account for. Also what level of genetic connectedness and confounding is present?*

**Response:** We accept the reviewer's point that this should be included. A Wald F-statistic table of fixed-effect significance for every trait, together with pedigree completeness and connectedness diagnostics, is now provided in Supplementary Material S1, and the confounding between breed and contemporary group is now described in the Results. To assess this, the variance of each breed proportion within contemporary groups was compared with its total variance. Texel and Suffolk proportions varied substantially within contemporary groups (within-group share of variance 0.64 and 0.68, respectively), as did Belclare to a lesser degree (0.40), whereas Cheviot (0.04) and Lleyn (0.15) proportions were largely nested within contemporary group, so their breed effects cannot be cleanly separated from contemporary-

group effects. The proportion of phenotypic variance accounted for by breed was calculated from each univariate model's breed-proportion solutions: it was below 1% for the methane traits recorded on all animals, and 0.2–10% for those recorded on the ADG and CT subsets, compared with 3% for MBW. As breed effects were estimated conditional on contemporary group, these values exclude any breed variation absorbed by the contemporary-group effects. Both sexes occurred across many contemporary groups, although 82.1% of individual contemporary groups contained a single sex. On reflection following the reviewer's comments, we also trimmed the pedigree to the direct ancestors of phenotyped animals only, as we believe this is more appropriate for a parameter-estimation study than a larger, sparser pedigree carrying many animals with no phenotyped descendants. [Revised manuscript, lines 186–191, 198, 284–289, 610–611, 617.]

*This led me to the equation on line 199 page 10 which I noted did not have a "+ e" at the end but is described in the text itself as if it were there.*

**Response:** Corrected; the model equation now explicitly includes the residual term (+ e). [Revised manuscript, line 194.]

*It appears that PAC data from different aged animals was combined in the analysis and these contemporary groups would have differed widely in liveweights and gas production, however, there was no discussion around scaling to normalise the variances of the contemporary groups nor even any discussion if this was examined.*

**Response:** This is correct, and we thank the reviewer for raising it. We have now tested directly whether a single combined analysis across growing and mature animals is appropriate. Residual variance heterogeneity across contemporary-group-mean classes was fitted for the methane traits with sufficient data spanning multiple stages; this is described in the Methods and Discussion. We also fitted a model in which residual variance differed by physiological stage but, on reflection, chose contemporary-group-mean classes instead, because residual variance

in methane traits tends to scale with the mean, and contemporary-group means vary widely within as well as between stages (e.g. with season, diet and measurement date). Grouping residuals by contemporary-group mean therefore captures this scale effect more fully, while still largely absorbing the stage difference, as contemporary groups are largely stage-specific. Where the heterogeneous-residual model converged (five of the nine methane traits), its heritability and repeatability estimates are now the ones reported in Table 2. For the reviewer's information, the corresponding heritabilities (SE) under a single homogeneous residual variance were 0.25 (0.02) for CH4, 0.19 (0.02) for CH4/MBW, 0.10 (0.01) for CH4 ratio, 0.18 (0.02) for RMTMBW and 0.18 (0.02) for RMTMBW+CO2; the genetic correlations remain from homogeneous-residual bivariate models, as a heterogeneous-residual bivariate model was not estimable. We also estimated the genetic correlation between methane production measured on growing animals and on mature animals, treated as two separate but genetically linked traits; this bivariate genetic correlation was 0.99, indicating that, genetically, methane production in young and old animals is essentially the same trait, which supports the appropriateness of the combined analysis. [Revised manuscript, lines 214–224, 516–530.]

*In summary, if this was just a genetic parameter estimation paper additional information would have been available to check if there was any underlying issues with the analysis and the values would have been formally compared and contrasted with other publications. This was not done, but the parameters estimated in at least one case, diverge markedly from previous published work and this has not been commented on. I suggest the analysis is re-examined and some additional results from the parameter estimation process are included at least in the supplemental material.*

**Response:** This is now addressed throughout: the discrepancy with Jonker et al. is discussed explicitly (see above and the Discussion section), and substantially more detail on how the genetic-parameter analysis was conducted is now given in the Methods and Supplementary

Material S1. Table 3 now reports the genetic and phenotypic correlations of CO2 and of each alternative definition with daily methane from the re-estimated models; the correlations among the alternative definitions themselves were not re-estimated and have been removed. [Revised manuscript, lines 284–285, 482–491, 610–611, 617.]

*The second issue relates to the selection of alternative methane trait definitions, including residual trait definitions. These on the surface appear arbitrary, but in fact, all they are doing are trying to correct for differences in feed intake using "proxy traits" as feed intake has not been measured. It is the amount of methane produced per unit feed intake and the amount of production per unit of feed intake that most closely relates to the question at hand "how can I maximise production of lamb carcass weight per hectare while reducing methane emissions" I feel this aspect has been not directly confronted, because there is an important additional caveat: around 70% of the feed is used in ewe maintenance and reproduction and only 30% in lamb carcass production. So increased fecundity, ewe longevity and lamb survival and associated traits relationship with methane traits are probably more important than liveweight or average daily gain in an enterprise's bio-economic analysis.*

**Response:** We accept the reviewer's point. The ratio and residual definitions are, in effect, proxies for feed-intake information that this dataset does not contain. The biological rationale for each definition, including the intake information it stands in for, is now set out in the Background and Discussion. CO2, which reflects total metabolic activity and is more directly linked to intake than growth rate, is now reported as a trait in its own right and is used, rather than ADG, as the intake proxy in the selection index. The Discussion also notes that the residual traits remain strongly genetically correlated with absolute methane (0.78–1.00), so they are not genetically independent of the traits they are adjusted for. We agree that a full bio-economic analysis would need to weigh feed intake per unit of production directly and that, as roughly 70% of ewe flock feed supports ewe maintenance and reproduction, fecundity, ewe longevity

and lamb survival are likely to be at least as important as the body-size and growth traits examined here. The selection index analysis is not intended as such a breeding objective: it includes no feed intake, fecundity, longevity or survival traits and no economic values. Its narrower aim, now stated in the Abstract, Background, Methods and Discussion, is to examine how absolute, ratio and residual methane traits behave within an index. It shows that a ratio or residual trait is, in effect, a fixed weighting of methane against body size, and that a favourable region, in which methane is reduced without a decline in MBW or CO2, exists (Figure 1) and can be reached from any of the methane traits once MBW and CO2 are weighted appropriately (Table 6). In line with the reviewer's point on feed cost, the reference index holds CO2 at zero response, so that methane is not reduced simply by favouring animals that eat less. [Revised manuscript, lines 237–240, 434–438, 465–479, 509–514, 539–542, 547–564.]

*The final issue relates to the use of the smith-hazel index theory used and this has two facets. The first is in several places the authors talk about progress per generation, but as far as I can understand from the paper (line 243, this actually equates to a selection intensity of i=1). I have no problem with this but would suggest the authors be more explicit throughout. In practice in sheep, the selection intensity per generation is likely to be closer to 1.7 and the generation length around 2.8 years. The second aspect that is not acknowledged in the paper is that in a real flock, not all animals may be measured for all traits, measurements may differ between sexes, and progeny, and genomics may or may not or only partially be used and if it is used the accuracy of the breeding values differ as well. The impacts of this are not explored here, and that is okay, but there does need to be acknowledgement along with references that these are real issues in practice and you cannot simply extrapolate from the graphs presented to a real world situation. The final aspect is that the use of Smith-Hazel index theory is at best an approximation and this is explored in Cuyabano, Beatriz C. D., Mariana R. Motta, Jeremie Vandenplas, Nancy L. Garcia, Fatima Shokor, Pascal Croiseau, Didier Boichard, Sophie*

*Aguerre, and Sophie Mattalia. 2025. "Trajectories of Genetic Correlations in Populations under Selection: From Theory to a Case-Study." bioRxiv. https://doi.org/10.1101/2025.03.13.643026. There needs to be some caveats and a reference to this work, as this issue is becoming more important with the use of genomics.*

**Response:** We thank the reviewer for these points, which we have addressed in the revised selection index analysis (Methods, "Selection index analysis"; Results, Figure 1 and Table 6; Supplementary Tables S2–S4). [Revised manuscript, lines 253–255, 273–282, 418–421, 547–553.]

1. Selection intensity: responses are now stated explicitly, in the Methods and in every table and figure caption, as per generation at a selection intensity of i = 1. Supplementary Table S2 gives the same responses at i = 1.7 and per year with a generation interval of 2.8 years (scaling by 1.7 and 0.61, respectively), as illustrative values.

2. Partial recording, sex, progeny and genomics: the analysis assumes selection on the candidate's own records for all three traits, with no relatives' or genomic information, no sex differences and no differences in breeding-value accuracy; we now state that these are limitations (lines 280–282, 547–553).

3. Approximation: we cite Cuyabano et al. (2025) [61] and state that the genetic parameters were treated as constant across generations, whereas they change under selection. We now propagate the sampling uncertainty of the parameter estimates into the index results by Monte Carlo simulation: the 15 (co)variance parameters of the trivariate model were resampled jointly (10,000 draws) from a multivariate normal distribution using their REML estimates and sampling (co)variance matrix, and results are reported with 95% intervals (Methods). We also state that the weights reported are specific to the parameters estimated in this population and should not be generalised.

4. Scope: we make clear throughout that the analysis is not intended as a large, sophisticated index simulation or a national breeding index. It is a deliberately narrow, detailed investigation of how methane traits behave within an index, and the simplifying assumptions above follow from that aim. Within this scope, we believe the conclusions on the relative behaviour of absolute, ratio and residual methane traits hold, even though the absolute responses should not be read as predictions for a real breeding programme.

## Reviewer #2

*The manuscript "Genetic Parameters and Selection Responses for Alternative Methane Trait Definitions in Pasture-Based Sheep" by Kerry et al. is very well written and the analysis provides a very comprehensive and relevant overview over the genetics of methane emissions in sheep. However, I have some general comments. You have a variety of methane phenotypes, however I miss a biological interpretation and possible effects you can conclude based on your results. Moreover, you could address in you methods-section why you decided to apply traits like e.g. the residual trait regressing methane on CO2 production. Why did you do that? As well, you obviously have the CO2 phenotype, it might be worth a consideration to include that trait in the analysis to be able to give some insights in the physiology and energy balance of the animals.*

**Response:** We thank the reviewer for this point and agree the original submission did not explain the biological rationale for the residual and ratio trait definitions, including regressing methane on CO2 production. We have kept the Methods section focused on describing the approach taken (how each trait was constructed), and have placed the biological rationale for these trait definitions in the Background and Discussion sections instead. We have also now included CO2 as a trait in its own right throughout the analysis: descriptive statistics, heritability and genetic standard deviation, and its genetic and phenotypic correlations with methane and metabolic body weight are all now reported, along with a biological interpretation

of what these relationships suggest about the animals' overall energy metabolism. CO2 is also included in the selection index analysis, both as a recorded trait and in the breeding goal; CH4 ratio, which when linearised gives the same breeding-goal direction as CH4/CO2, is among the methane traits examined in the index (Table 6). We use CO2 as a proxy for feed intake, while recognising that it is not a validated measure of intake. [Revised manuscript, lines 165–166, 323–325, 354–355, 465–479, 509–514.]

*Regarding the PAC method - what are pros and cons regarding you study? Does that capture the real emission of the animals? And how do you address the in general very little amount of repeatable records per animal?*

**Response:** The pros and cons of the PAC method for this study are now discussed in the Background. In brief, PACs are portable, quick (about one hour per measurement) and relatively low cost per phenotype, so large numbers of animals can be measured on farm under commercial pasture-based conditions, which is what makes a genetic evaluation of this size feasible; PAC measurements are also repeatable and related to respiration-chamber measurements. Their main limitation is that each record is a short snapshot taken off pasture, rather than a full 24-hour measurement, so it is sensitive to time since feeding and diurnal variation, and it does not capture feed intake directly. PAC records therefore rank animals for methane output, but do not give an exact daily emission for each animal. On the reviewer's point about the generally small number of repeated records per animal: we acknowledge this as a real limitation of the dataset and now state this explicitly. At the same time, we believe it remains important to account for a permanent environmental effect in the model even with a modest proportion of repeated records, given the amount of occasion-to-occasion variability observed in methane measurements; omitting it would risk attributing repeatable but non-genetic variation to the additive genetic variance. We report repeatability alongside heritability for every trait for which the permanent environmental variance could be estimated and note

directly in the results where it could not (itself a further indication of how sparse repeated records limit what can be estimated for some of the smaller trait subsets, such as the CT-derived traits). [Revised manuscript, lines 66–73, 530–532.]

*I would also like you to mention in the discussion-section how you findings relate to other study´s findings. Are they somewhat similar? Or are there any surprises? And as you mention - Ireland is a grass-based system. Are your results transferable to other production systems?*

**Response:** We agree. The Discussion now compares our heritability estimates with published estimates for daily methane, CH4 ratio, methane relative to product and residual methane in sheep, beef and dairy cattle. The heritability of daily methane matched PAC estimates in New Zealand lambs and ewes (0.19–0.20; Jonker et al., 2018) and was lower than the sheep respiration-chamber estimate (0.29). In cattle, the highest estimates come from GreenFeed measurements averaged over a full test period, consistent with a single short PAC measurement capturing less of an animal's average emission. The CH4 ratio heritability now falls within the 0.17–0.25 range reported by Jonker et al. (2018). Methane relative to product (0.21–0.66) and residual methane (0.38–0.46) were generally more heritable in cattle than the corresponding traits here, which adjust for body size, growth and CO2 as proxies for intake rather than for measured intake or output. On transferability, our data come from Irish pasture-based flocks of several breeds, recorded at grazing in lambs, hoggets and ewes. Heritabilities for daily methane and CH4 ratio were similar to those of PAC-measured grazing sheep in New Zealand and Australia, and sire rankings for PAC methane have been maintained between grazing and controlled environments (Wahinya et al., 2022). This suggests the parameters apply to other grazing sheep systems, while transfer to housed or concentrate-fed systems, where diet and intake differ, remains to be tested. [Revised manuscript, lines 482–491.]

*At last - there are other studies showing differences between growing and adult animals in their physiology and therefore also in their methane emissions. Did you consider splitting the dataset into different groups? That might affect your results.*

**Response:** We thank the reviewer for this point. As described in our response to Reviewer 1 above, we now formally test this through the heterogeneous-residual-variance analysis and the genetic correlation between methane production in growing and mature animals (0.99). Together, these support a combined analysis for this dataset and directly answer the reviewer's question about whether growing and adult animals differ physiologically in their methane emissions. [Revised manuscript, lines 214–224, 516–530.]

Some minor comments are listed below:

***Line 47:*** *I would maybe mention that this is the ratio of these traits - looks a bit better to read.*

**Response:** The Abstract has been rewritten, and the sentence on selection for CH4/MBW has been removed. The revised Abstract now states explicitly that the alternative traits are ratios of methane relative to metabolic body weight, average daily gain, muscle mass and rumen volume, or residuals of methane regressed on these traits. [Revised manuscript, lines 26–28.]

***Line 48 ff:*** *I would recommend to write g/day instead, that looks a bit better to read.*

**Response:** We thank the reviewer for this suggestion. We have retained the notation g $day^{-1}$, as it follows SI convention and remains unambiguous for the compound units used throughout the manuscript (e.g. g $day^{-1}$ $kg^{-0.75}$ for CH4/MBW and g $day^{-1}$ $L^{-1}$ for CH4/rumen), which would be harder to read with slashes. We are happy to change this if the journal style requires.

***Line 70-71:*** *I would maybe rephrase "climate mitigation" - you want to mitigate greenhouse gas emissions, not the climate I guess..*

**Response:** “climate mitigation strategies” now reads “greenhouse gas mitigation strategies” (Background). [Revised manuscript, lines 46–47.]

***Line 72:*** *What do you mean with agriculture dominated by methane?*

**Response:** The sentence now states that agriculture is the largest single sector source of Ireland's national greenhouse gas emissions, with methane from enteric fermentation in cattle and sheep the dominant gas (Background). [Revised manuscript, lines 54–56.]

***Line 73:*** *How is that? I would recommend you to explain this statement, or to remove it in order to avoid confusion.*

**Response:** The statement on Ireland's carbon efficiency is now explained as being on a per-kilogram-of-product basis and cited [5, 6], and is linked to the need for mitigation that does not reduce productivity (Background). [Revised manuscript, lines 56–58.]

***Line 84-85:*** *I would generally recommend you do define in the beginning what you mean with methane production, methane emission and so on. Methane emissions can also include those from manure, but as you mean the ones from enteric fermentation I would recommend you to specify this in the beginning.*

**Response:** Enteric methane is now specified at first use in the Background ("Enteric methane emissions (hereafter referred to as methane emissions), produced during rumen fermentation…"), so that methane from manure management is not implied. The Methods (Absolute Methane Production) restates this as the methane captured by PAC measurement. [Revised manuscript, lines 47–48, 163–164.]

***Line 154:*** *Charolais with only one "l".*

**Response:** Corrected to "Charolais" throughout; the misspelling "Llyen" in the same sentence has also been corrected to "Lleyn". [Revised manuscript, lines 143–144.]

***Line 164:*** *You consistently call these traits "alternative" definitions, but I would recommend you to call them "different" or something similar, as I always remain asking myself alternative to what?*

**Response:** We have kept the term "alternative", as it is widely used for these traits in the literature, but it is now defined where the traits are introduced in the Methods (Methane trait

definitions): “alternative” refers to traits expressed relative to production or as residuals, rather than as daily (absolute) methane alone. [Revised manuscript, lines 158–161.]

***Line 195:*** *How many sires did you have and how complete was the pedigree for all animals?*

**Response:** The Methods (Genetic analysis) now state the pedigree size (36,449 animals), the number of unique sires (1,025), pedigree completeness (100% of phenotyped animals in the pedigree; 88.5% with both parents known), mean pedigree depth (17.6 generations) and sire connectedness (73.1% of sires linked two or more contemporary groups). [Revised manuscript, lines 186–191, 617.]

***Line 202 ff:*** *"sex" and the other effects in the model should also be cursive when referring to the model.*

**Response:** All model terms (sex, BR to TX, UN, het, rec, age, BTg, RTg, BTe, RTe, DP, CG, a, pe and e) are now italicised in the model definitions (Methods, Genetic analysis). [Revised manuscript, lines 195–211.]

***Line 213:*** *You stated initially that only about 25% of you animals have repeated records and though, you apply a permanent environmental effect. Did you other techniques?*

**Response:** See our response above on repeated records. Given the substantial occasion-to-occasion variability of single short PAC measurements, we did not consider a model without the permanent environmental effect to be appropriate, even with about a quarter of animals having repeated records. Omitting it would attribute repeatable but non-genetic variation to the additive genetic variance, inflating heritability, and would not allow repeatability to be estimated, which is itself an important parameter for PAC-based methane phenotypes. The animal permanent environmental effect is therefore retained in the model, and repeatability is reported alongside heritability (Table 2). The limited number of repeated records per animal is now stated as a limitation. [Revised manuscript, lines 530–532.]

***Line 297:*** *You did not introduce the CH4/CO2 ratio as a trait in your method section.*

**Response:** The trait at line 297 is CH4 ratio, which is CH4/(CH4 + CO2) rather than CH4/CO2. It was defined in the submitted Methods (Ratio-Based Methane Traits), but we have now made the definition more explicit there, stating the formula, its units (g day$^{-1}$ for both gases) and that it is calculated on a mass basis. [Revised manuscript, lines 170–171.]

***Line 304:*** *Why are there two numbers for CH4 ratio?*

**Response:** In the submitted Table 2, the standard errors of σa and σpe for CH4 ratio ($1.23\times10^{-4}$ and $2.90\times10^{-5}$) wrapped onto a second line, which made the row appear to give two sets of values. Table 2 has been reformatted so that each trait occupies a single row, and CVa is no longer reported for CH4 ratio. The manuscript now reports a single heritability and repeatability for CH4 ratio, 0.19 and 0.32, from the model with residual variance allowed to differ across contemporary-group-mean classes (Table 2). For the reviewer's information, the estimate under a single homogeneous residual variance was 0.10 ± 0.01. [Revised manuscript, lines 339–341.]

***Line 304:*** *I would like you to introduce how you calculated repeateabilities and heritabilities in the methods section. Just for completeness of this section.*

**Response:** The formulas are now given as displayed equations in the Methods (Genetic analysis): $h^2 = \sigma^2_a / (\sigma^2_a + \sigma^2_{pe} + \sigma^2_e)$ and $t = (\sigma^2_a + \sigma^2_{pe}) / (\sigma^2_a + \sigma^2_{pe} + \sigma^2_e)$, with the terms as defined for the model. [Revised manuscript, lines 228–230.]

***Line 499:*** *This might also be because you have a phenotypical residual methane trait, not one that is genetically independent of the traits it is regressed on. Might be noteworthy to address that.*

**Response:** The Discussion now notes that the residual traits remain strongly genetically correlated with absolute methane (rg = 0.78–1.00), consistent with the general statistical behaviour of residual variables; that is, a phenotypic residual is not genetically independent of the traits it is regressed on. The index analysis adds that a residual is a fixed weighting of

methane against the adjustment traits (Discussion). [Revised manuscript, lines 506–508, 539–542.]